is proportional to the potential difference between the plates.

$$Q \propto V$$

so that Q = CV

where the C is the proportionality constant called capacitance. **The capacitance C of a capacitor is defined as the ratio of the magnitude of charge on either of the conductor plates to the potential difference existing between the conductors.**

$$C = \frac{Q}{V} \qquad (1.81)$$

The SI unit of capacitance is c*oulomb per volt* or *farad (F)* in honor of Michael Faraday. Farad is a very large unit of capacitance. In practice, capacitors are available in the range of microfarad (1μF = $10^{-6}$ F) to picofarad (1pf = $10^{-12}$ F). A capacitor is represented by the symbol ⊣⊢ or ⊣( . Note that the total charge stored in the capacitor is zero (Q - Q = 0). When we say the capacitor stores charges, it means the amount of charge that can be stored in any one of the plates.

Nowadays there are capacitors available in various shapes (cylindrical, disk) and types (tantalum, ceramic and electrolytic), as shown in Figure 1.55. These capacitors are extensively used in various kinds of electronic circuits.

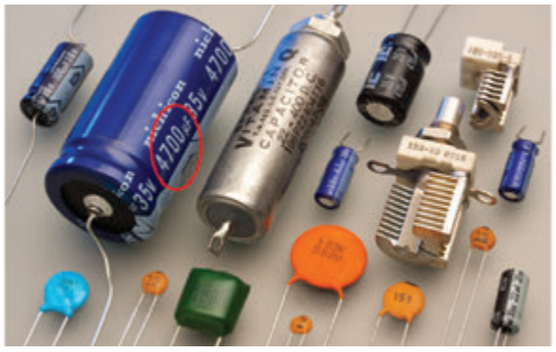


**Figure 1.55** Various types of capacitors

**Capacitance of a parallel plate capacitor**

Consider a capacitor with two parallel plates each of cross-sectional area A and separated by a distance d as shown in Figure 1.56.

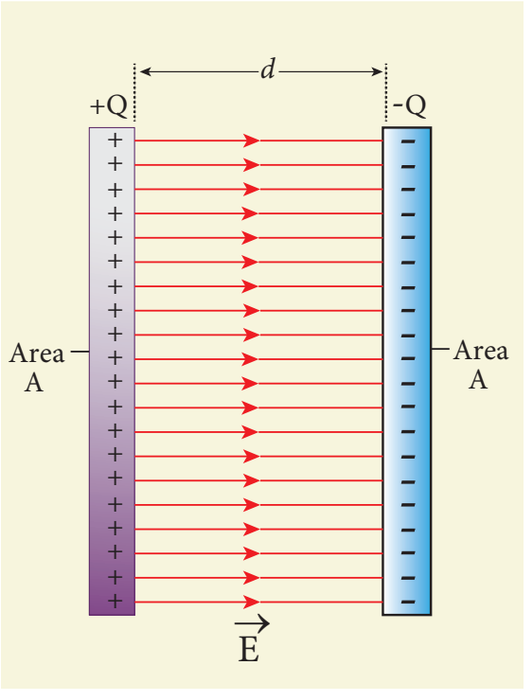



**Figure 1.56** Capacitance of a parallel plate capacitor

The electric field between two infinite parallel plates is uniform and is given by $E = \frac{\sigma}{\varepsilon_0}$ where σ is the surface charge density on the plates $\left(\sigma = \frac{Q}{A}\right)$. If the separation distance d is very much smaller than the size of the plate ($d^2 << A$), then the above result is used even for finite–sized parallel plate capacitor.

The electric field between the plates is

$$E = \frac{Q}{A\varepsilon_0} \qquad (1.82)$$

Since the electric field is uniform, the electric potential between the plates having separation d is given by



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